

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: _____Gan Hounan_____

Student ID: _____22097776d_____

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 128

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

This information indicates that R2 only established an OSPF adjacency with R1 (via Serial0/0/0) and not with R3, because only R2's Serial0/0/0 was re-enabled for OSPF (no passive-interface), while Serial0/0/1 remained a passive interface. As a result, R3 has no direct OSPF neighbor relationship with R2 and must forward traffic to the 192.168.2.0/24 network through R1 (via R3's Serial0/0/0), leading to an accumulated cost metric of 128 (the sum of two serial link OSPF costs: 64 + 64).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

~~64~~, It is calculated as the default OSPF cost of a single serial link (R3 S0/0/1 directly connected to R2 S0/0/1)

Is R2 listed as an OSPF neighbor to R3? ___Yes___

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default OSPF reference bandwidth is 100 Mbps. All links with a bandwidth of 100 Mbps or higher (e.g., Gigabit Ethernet, 10 Gigabit Ethernet) will have an OSPF cost of 1, which cannot distinguish the bandwidth differences of high-speed links. Changing the reference bandwidth (e.g., to 10000 Mbps) allows different high-speed links to generate different cost values, making OSPF route selection more consistent with the actual link bandwidth and improving the accuracy of route planning.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: The total cost is 782, calculated as the cost of R1 S0/0/1 (781) plus the default cost of R3 G0/0 (1), i.e., $781 + 1 = 782$.

192.168.23.0/30: The total cost is 845, calculated as the cost of R1 S0/0/1 (781) plus the default cost of R3 S0/0/1 (64), i.e., $781 + 64 = 845$. (The same cost applies to the path via R1 S0/0/0 and R2 S0/0/1.)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost is 1562. Because the bandwidth of all serial interfaces is set to 128 Kbps, each serial interface has an OSPF cost of 781. There are two equal-cost routes from R1 to 192.168.23.0/30, both passing through two serial links (R1 S0/0/0 → R2 S0/0/1 or R1 S0/0/1 → R3 S0/0/1). The total cost is $781 + 781 = 1562$ for both paths.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The S0/0/1 interface of R1 is manually configured with `ip ospf cost 1565`, which is much higher than the accumulated cost of the path to 192.168.3.0/24 through R2 (1563). The accumulated cost of the path through

R2 is calculated as the cost of R1 S0/0/0 (781) plus the cost of R2 S0/0/1 (781) plus the default cost of R3 G0/0 (1), i.e., $781+781+1=1563$. OSPF uses the Shortest Path First (SPF) algorithm and always selects the route with the lowest cost, so R1 chooses the link through R2 to access 192.168.3.0/24.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The OSPF router ID is a unique identifier for routers in the OSPF domain, which is used for neighbor adjacency establishment, LSA generation and exchange, and SPF algorithm calculation. A router ID conflict will prevent the establishment of OSPF adjacencies and cause abnormal exchange of routing information. Manually controlling the router ID makes network planning clearer and facilitates troubleshooting (e.g., quickly locating routers by router ID). It also prevents the automatically selected router ID (the highest IP address of physical interfaces) from changing due to interface failures or IP address modifications, which avoids re-convergence of the OSPF topology and improves network stability.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR election only occurs in OSPF **Broadcast** and **NBMA (Non-Broadcast Multi-Access)** network types. In this lab, the OSPF network type of serial links is **Point-to-Point**, which does not require DR/BDR election. Only the Ethernet interfaces (G0/0) are set to Broadcast network type, but each Ethernet interface is only connected to a single PC with no other OSPF routers, so DR/BDR election is not necessary.

3. Why would you want to set an OSPF interface to passive?

A passive OSPF interface suppresses the sending and receiving of OSPF Hello packets while still advertising the network of the interface into the OSPF domain. This setting avoids unnecessary OSPF protocol traffic on LAN segments (e.g., G0/0 connected to PCs), reducing router resource usage and network bandwidth consumption. It also prevents irrelevant devices on the LAN from receiving OSPF routing information, improving network security. In addition, it avoids invalid adjacency negotiation on interfaces with no OSPF neighbors, simplifying the OSPF topology.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The **Area partitioning** mechanism is the core of OSPF's hierarchical design and the key to its high scalability. A large OSPF domain can be divided into multiple areas, all of which are connected to the **backbone area (Area 0)**. Routing information between areas is only exchanged by ABRs (Area Border Routers) in the form of summarized routes. Routers in each area only maintain the topological information of their own area, and LSAs are only flooded within the area, which reduces the size of the routing table and the computational load of the SPF algorithm on each router. The backbone area is responsible for inter-area route forwarding, ensuring the connectivity of the entire OSPF domain, and route summarization further reduces the propagation of routing information.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

Cisco routers prioritize **Administrative Distance (AD)** over metric when selecting routes. The default AD values are: 120 for RIP, 110 for OSPF, and 90 for internal EIGRP routes. A smaller administrative distance indicates a more reliable routing protocol, so the router will select the EIGRP route regardless of its metric value.